Appendix A. The Number Systems

Natural numbers, integers, rationals, and the construction of the reals by Dedekind cuts

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Chapter 1 took the real numbers as a complete ordered field. Here we sketch how such a field is built from the natural numbers, so that the axioms are consistent.

A.1 Natural numbers and induction

The natural numbers are characterized by Peano's axioms: there is a set $\mathbb{N}$ with an element 1 and an injective successor map $s : \mathbb{N} \to \mathbb{N}$ with $1 \notin s(\mathbb{N})$, such that any subset containing 1 and closed under s is all of $\mathbb{N}$ (**induction**). Addition and multiplication are defined recursively ($m+1 = s(m)$, $m + s(n) = s(m + n)$; $m \cdot 1 = m$, $m \cdot s(n) = mn + m$), and their associative, commutative and distributive laws, as well as the order $m < n \iff n = m + k$ for some k, are proved by induction. Induction is equivalent to the **well-ordering principle**: every nonempty subset of $\mathbb{N}$ has a least element.

A.2 Integers and rationals

$\mathbb{Z}$ is obtained from pairs $(m, n) \in \mathbb{N}\times\mathbb{N}$ (thought of as $m-n$) modulo $(m, n) \sim (m', n') \iff m+n' = m'+n$, with $(m, n)+(m', n') = (m+m', n+n')$ and $(m, n)(m', n') = (mm'+nn', mn'+nm')$; these operations are well defined on classes, and $\mathbb{Z}$ is a commutative ring with an order. $\mathbb{Q}$ is obtained from pairs $(p, q) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ (thought of as p/q) modulo $(p, q) \sim (p', q') \iff pq' = p'q$; it is an ordered field (positive elements: classes with $pq > 0$) which is Archimedean but not complete.

A.3 Real numbers by Dedekind cuts

Definition (Cut)

A **cut** is a subset $\alpha \subset \mathbb{Q}$ such that $\emptyset \neq \alpha \neq \mathbb{Q}$; if $p \in \alpha$ and $q < p$ then $q \in \alpha$; and α has no largest element. $\mathbb{R}$ is the set of cuts, ordered by inclusion.

Each rational r gives the cut $r^* = \{p \in \mathbb{Q} : p < r\}$, and $\{p \in \mathbb{Q} : p < 0\} \cup \{p \geq 0 : p^2 < 2\}$ is a cut not of this form.

Theorem A.1

(a) The order on $\mathbb{R}$ is total, and every nonempty set of cuts bounded above has a supremum, namely its union. (b) With $\alpha + \beta = \{p + q : p \in \alpha, q \in \beta\}$ and, for positive cuts, $\alpha\beta = \{pq : p \in \alpha, q \in \beta, p, q > 0\} \cup \{p \leq 0\}$ (extended to other signs by the usual rules), $\mathbb{R}$ is an ordered field. (c) $r \mapsto r^*$ embeds $\mathbb{Q}$ into $\mathbb{R}$ as an ordered subfield.

Proof. (a) If $\alpha \not\subset \beta$, pick $p \in \alpha \setminus \beta$; every $q \in \beta$ satisfies $q < p$ (otherwise $p \in \beta$), so $\beta \subset \alpha$. For A bounded by a cut β, the union $\gamma = \bigcup A$ is a cut ($\gamma \subset \beta \neq \mathbb{Q}$; downward closed; no maximum since no member has one), and it is clearly the least upper bound. (b) Most axioms are routine; the key points are the additive inverse $-\alpha = \{p : -p - r \notin \alpha$ for some rational $r > 0\}$ and the identity $\alpha + (-\alpha) = 0^*$, which uses the Archimedean property of $\mathbb{Q}$ to find $p \in \alpha$ with $p + w \notin \alpha$ for a given small $w > 0$. Multiplicative inverses are handled similarly for positive cuts. (c) The map preserves sums, products and order. ∎

Thus a complete ordered field exists, and by the uniqueness result of Chapter 1 it is unique up to isomorphism. Other constructions (equivalence classes of Cauchy sequences of rationals, or infinite decimals) lead to the same field. The complex numbers are then $\mathbb{R}^2$ with $(a, b)(c, d) = (ac - bd, ad + bc)$.